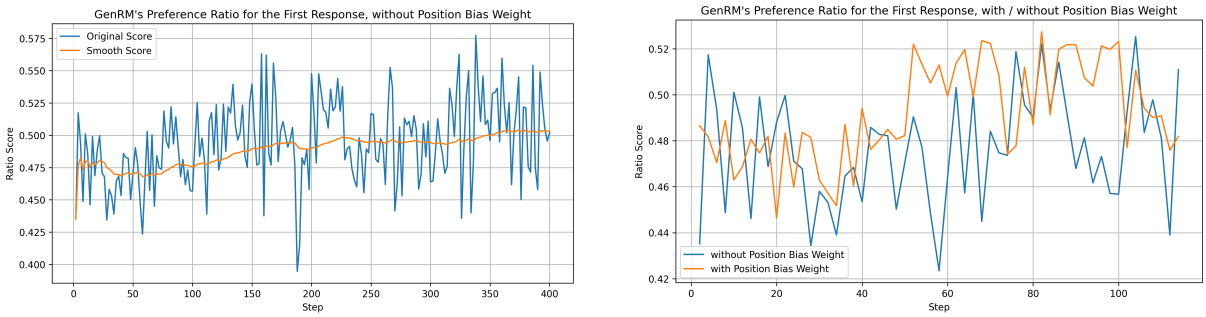
Model	Mean Response Len	Mean Explanation Len
Qwen3-32B-Base-ScalarRM-GRPO	1872	417
Qwen3-32B-Base-GenRM-BRPO / **Writing-Zero**	1292	58
Writing-SFT	1251	125

Table 3: Mean length of response and redundant explanation.

quantify these effects, we analyzed the performance of three key models on these hacking issues using an internal creative writing test set. As shown in Table 3, our **Writing-Zero** model demonstrates substantially less redundant information and shorter response lengths compared to `Qwen3-32B-Base-ScalarRM-GRPO`, highlighting its improved resistance to these hacking phenomena.

5.2 GenRM's Position Bias



(a) Preference Ratio without Position Bias Weight

(b) Preference Ratio with Position Bias Weight

Figure 4: Convergence of the GenRM's preference ratio during RL training.

After the cold-start fine-tuning (RFT) detailed in Section 3.1.3, our initial fine-tuned GenRM exhibited a significant position bias. Unlike Claude-3.5-Sonnet, which showed a 60% tendency to favor the first response, our model initially preferred to assign higher scores to the latter response in a pair. Fortunately, this inherent bias was largely calibrated during the subsequent RL phase. As illustrated in Figure 4a, the empirical probability of the GenRM preferring the former response gradually converged towards the ideal 50% during RL training.

While the mean of this preference probability (as shown in Figure 4a) converged towards 0.5, its variance across training batches did not show a similar convergent tendency without intervention. As demonstrated in Figure 4b, the position bias advantage weighting mechanism visibly reduced the variance of the preference ratio.

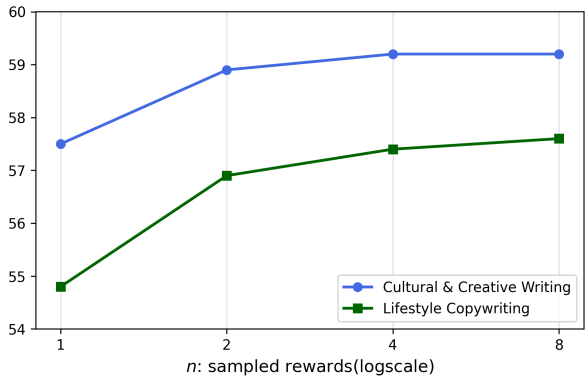


Figure 5: Majority voting accuracy (voting@n) across $n = 1, 2, 4, 8$ on internal datasets, as evaluated by the pairwise GenRM.

5.3 Test-time Scaling Benefits

A notable advantage of GenRMs over scalar reward models is their capability for test-time scaling through majority voting over multiple generations. This can enhance both the direct accuracy of the GenRM itself and, consequently, the performance of the policy model trained with it.

As illustrated in Figure 5, increasing the number of voting samples (n) directly improves the accuracy of our Pairwise Writing GenRM on our internal evaluation datasets. For instance, with $n = 8$ votes, accuracy on the Cultural & Creative Writing dataset increases from 57.5% (for $n = 1$) to 59.2%, and on the Lifestyle Copywriting dataset from 54.8% to 57.6%. This demonstrates the GenRM's ability to refine its judgments by aggregating multiple perspectives. The benefits of test-time scaling extend to the policy models trained with the GenRM. As shown in Table 2, when **Writing-Zero** utilizes its Pairwise GenRM with two voting rounds (voting@2), it achieves further improvements, which indicates that a more accurate reward signal, refined by test-time scaling, can lead to better policy optimization and ultimately stronger performance on downstream writing tasks. Further enhancing the GenRM's performance could unlock greater test-time scaling benefits, for example, by enabling strategies for multiple voting rounds (n ¿ 1) that selectively reward consistently superior responses while neutralizing or penalizing inconsistently judged ones, thereby fostering more robust policy learning.

We believe that with further improvements to the GenRM's accuracy, the potential of test-time scaling with multiple voting rounds could be more deeply explored. For instance, an open question for future experimentation is whether responses with inconsistent judgments across multiple votes should be neutrally rewarded (e.g., 0) or negatively penalized (e.g., -1 in Equation 11).

5.4 Intuition behind reference selection of BRPO

The primary challenge in non-verifiable tasks like creative writing is the absence of definitive ground-truth answers, making quality assessment inherently relative and reliant on comparison. We initially considered using the best-performing or relatively better response from a previous group rollout as reference for the next training iteration. However, this approach proved problematic. As the policy model's capabilities significantly improve during training, such a static or outdated reference quickly becomes suboptimal, introducing an "offline" data issue that can impede further learning by comparing against a weaker baseline. Furthermore, our pairwise GenRM is designed for relative comparisons rather than assigning absolute scores. Attempting to derive a consistent listwise ranking from purely pairwise preferences would necessitate an additional redundant pointwise reward model. Consequently, BRPO adopts a dynamic, bootstrapped reference selection strategy, where a randomly chosen response from the current policy rollouts serves as a temporary, relevant reference for pairwise comparisons within the same group, fostering continuous self-improvement without external anchors.

6 Related Work

6.1 Reward Model

Early approaches for reward modeling primarily focused on discriminative reward models, which were trained to directly assess the quality of individual responses (Liu et al., 2024; Cai et al., 2024). These pointwise assessment methods laid the groundwork for preference learning, but frequently faced challenges with generalization across diverse contexts and over-optimization (Yang et al., 2024; Sharma et al., 2023; McKee-Reid et al., 2024). Another line of work gives pairwise scores to evaluate whether responses match the reference answer, or compare the quality of two responses (Xu et al., 2025; Jiang et al., 2023), showing that by learning to predict which of two responses is preferred, generally leads to improved stability and better capture the relative nuances of human preferences compared to direct scalar assessments. Generative Reward Models (GenRMs) introduce a new paradigm (Li et al., 2023; Vu et al., 2024; Ye et al., 2024; Zhang et al., 2024; Cao et al., 2024), which allows the reward model to leverage chain-of-thought and even use test-time scaling to make more reliable evaluation. Liu et al. (2025) stand out as the most relevant works to our work. They introduced a novel learning method enabling GenRMs to generate adaptive principles and accurate critiques to give a more reliable comparison between two responses through a two-phase training process: distillation then RLVR. Chen et al. (2025a); Whitehouse et al. (2025); Chen et al. (2025b) share a similar idea of using designed evaluation criteria and RLVR to improve the quality of GenRM. Zhang et al. (2025); Wang et al. (2025) also expand GenRM and RLVR to multimodal domains. In this work, we explore the application of pairwise GenRM to non-verifiable writing tasks.

6.2 Reinforcement Learning

Reinforcement Learning from Human Feedback (RLHF) has become a crucial component for aligning Large Language Models (LLMs) with human preferences and desired behaviors (Ouyang et al., 2022). Recently, Reinforcement Learning with Verifiable Rewards (RLVR) has proven an effective paradigm for improving the reasoning capabilities of LLMs in domain such as math and coding (Jaech et al., 2024; Guo et al., 2025; Team et al., 2025), inspiring subsequent research to actively explore various potentially more advanced reasoning RL algorithms (Yu et al., 2025; Xiong et al., 2025; Li et al., 2025). Su et al. (2025) and Xu et al. (2025) propose to expand RLVR to problems with unstructured reference, which often rely on a pre-defined fixed reference or a best-of-n responses sampled from an initial or static model, to serve as a ground-truth reference. In this work, our proposed Bootstrapped Relative Policy Optimization (BRPO) algorithm is a step further to explore the application of RLVR to non-verifiable tasks.

7 Conclusion, Limitations, and Future Work

In this work, we propose a unified RLVR-based training paradigm to bridge the gap between non-verifiable tasks and verifiable rewards. By introducing a writing-principle-based pairwise Generative Reward Model and the Bootstrapped Relative Policy Optimization algorithm, our approach enables large language models to achieve stable and scalable reinforcement learning, even in highly subjective domains such as creative writing. Experimental results demonstrate that our method not only enhances robustness against reward hacking but also yields competitive performance across different writing benchmarks. This work lays the foundation for a comprehensive and consistent RL framework that is applicable across the full spectrum of language tasks. However, due to limited computational resources, we were unable to systematically analyze the trends of test-time scaling of our pairwise writing GenRM, nor could we fully investigate the impact of various parameters and settings within both the GenRM and RL stages. We also leave the integration of all reward modeling paradigms in a unified framework—along with the extension to multi-modal scenarios—as directions for future research.

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